

1. IDS, IPS, and Firewall Concepts

Intrusion Detection System (IDS)

- **Definition:** A software system or hardware device that inspects all inbound and outbound network traffic for suspicious patterns indicating a security breach. It alerts security personnel upon detection.
- **Main Functions:**
 - Gathers and analyzes information to identify policy violations and misuse.
 - Acts as a "packet sniffer," intercepting and analyzing TCP/IP packets.
 - Evaluates traffic for suspected intrusions and raises alarms.
- **Placement:** Can be placed outside/inside the firewall. Best practice is layered defense (before and after the firewall).
- **How it Works:** Uses sensors to detect malicious signatures or behavioral anomalies. If a match occurs, it performs predefined actions (terminate connection, block IP, drop packet, alarm).
- **Types:**
 - **Passive IDS:** Only detects intrusions and alerts.
 - **Active IDS (IPS):** Detects and also prevents intrusions.

Intrusion Prevention System (IPS)

- **Definition:** An active IDS placed *inline* in the network (between source and destination) to actively analyze traffic and make automated decisions to prevent intrusions.
- **Actions:** Generates alerts, records real-time logs, blocks/filters malicious traffic, detects/eliminates threats quickly, identifies threats accurately.
- **Advantages over IDS:** Can actively block/drop illegal packets, monitor activities within an organization, prevent direct attacks by controlling traffic.
- **Classification:**
 - **Host-based IPS (HIPS):** Protects a single host.
 - **Network-based IPS (NIPS):** Protects network segments.

How an IDS Detects an Intrusion

1. **Signature Recognition (Misuse Detection):**
 - Compares incoming traffic with known attack signatures (binary patterns of known attacks).
 - Detects known attacks but can generate false positives if innocuous packets match signatures.
 - Requires a massive number of signatures, which can impact performance.

2. Anomaly Detection:

- Detects intrusions based on deviations from established "normal" behavioral characteristics of users and system components.
- Builds a baseline of normal activity; anything outside this threshold is an anomaly.
- Challenging to create precise models due to network traffic unpredictability.

3. Protocol Anomaly Detection:

- Analyzes network traffic to detect deviations from established protocol standards or expected behavior patterns (e.g., unusual packet structures, unexpected sequence orders).
- Assumes most protocols have defined rules; violations indicate malicious activity or misconfiguration

General Indications of Intrusions

- **File System Intrusions:** Modification/deletion of files, unauthorized access, unexplained file changes, rogue SUID/SGID files, missing files, unexplained disk usage, abnormal system behavior.
- **Network Intrusions:** Sudden increase in bandwidth consumption, repeated probes, connection requests from unauthorized IPs, repeated login attempts, sudden influx of log data, unexpected network configuration changes.
- **System Intrusions:** Sudden changes in logs, slow performance, missing logs/permissions, disabled security software, unusual graphic displays, system crashes/reboots, unfamiliar processes, presence of attacker tools.

Types of Intrusion Detection Systems

Network-Based Intrusion Detection Systems

- These systems typically consist of a black box that is placed on the network in a promiscuous mode, listening for patterns indicative of an intrusion
- It detects malicious activity such as Denial-of-Service attacks, port scans, or even attempts to crack into computers by monitoring network traffic

Host-Based Intrusion Detection Systems

- These systems usually include auditing for events that occur on a specific host
- These are not as common, due to the overhead they incur by having to monitor each system event

Types of IDS Alerts

- **True Positive (Attack - Alert):** An actual attack occurs and the IDS correctly raises an alarm.
- **False Positive (No Attack - Alert):** No actual attack, but the IDS incorrectly raises an alarm.
- **False Negative (Attack - No Alert):** An actual attack occurs, but the IDS fails to raise an alarm (most dangerous).
- **True Negative (No Attack - No Alert):** No attack occurs and the IDS correctly remains silent.

Firewall

- **Definition:** A software or hardware-based system that protects network resources by filtering incoming and outgoing traffic based on predefined security rules.
- They are placed at the junction or gateway between two networks, which is usually between a private network and a public network such as the Internet
- **Function:** Examines all messages entering/leaving a private network and blocks those that do not meet specific criteria.
- **Key Capabilities:**
 - Intrusion detection mechanism.
 - Filters packets based on address, type, port numbers.
 - Recognizes source/destination addresses.
 - Identifies network traffic type and data packet attributes.
 - Logging all attempts for auditing.

Firewall Architecture

Bastion Host

• A bastion host is a computer system designed and configured to protect network resources from attacks

1. a public interface directly connected to the Internet
2. a private interface connected to the Intranet

Screened Subnet

• The screened subnet or Demilitarized Zone (DMZ) contains hosts that offer public services • The DMZ responds to public requests, and has no hosts accessed by the private network • This private zone can not be accessed by Internet users

Multi-homed Firewall

• In this case, a firewall with two or more interfaces is present that allows further subdivision of the network based on the specific security objectives of the organization

DMZ (Demilitarized Zone)

→ DMZ is a network that serves as a buffer b/w the internal secure network and insecure internet.

→ It can be created with Firewalls with 3 or more interfaces.

Types of Firewalls

Firewalls protect networks by monitoring and filtering traffic. They are categorized by **configuration** and **working mechanism**.

I. Based on Configuration

1. Network-Based Firewalls

- Placed at the **network perimeter** to inspect packets.
- Protects LANs by filtering traffic based on headers.

Pros: Centralized control, handles high traffic

Cons: Expensive, complex to set up e.g., Cisco Secure Firewall ASA, FortiGate, Check Point Quantum Security Gateway

2. Host-Based Firewalls

- Installed on **individual systems** as software.
- Filters traffic to/from that host.

Pros: Cost-effective, good for endpoint protection

Cons: Hard to manage at scale, uses system resources e.g., Microsoft Defender Firewall, ZoneAlarm, Comodo Firewall

II. Based on Working Mechanism

1. Packet Filtering Firewall

- Filters by **IP, port, and protocol** at the network layer.

Pros: Fast, low overhead

Cons: No session awareness, vulnerable to spoofing

2. Circuit-Level Gateway

- Operates at the **session layer**, checks TCP handshakes.

Pros: Hides internal structure, low resource use

Cons: Doesn't inspect packet content

3. Application-Level Firewall (Proxy)

- Filters by **application protocols** (HTTP, FTP, etc.).

Pros: Deep inspection, service-specific rules

Cons: Slower, complex setup

4. Stateful Multilayer Inspection

- Tracks **sessions** and inspects packets at multiple layers.

Pros: Combines filtering, session, and app-level checks

Cons: Resource-intensive, complex

5. Application Proxy

- Acts as a **middleman**, filtering based on specific services.

Pros: App-specific control, better privacy

Cons: Slower, limited to supported apps

6. VPN Firewall

- Secures traffic in **VPN tunnels** with encryption + filtering.

Pros: Encrypts data, filters remote traffic

Cons: Vulnerable if endpoints are compromised

7. Next-Generation Firewall (NGFW)

- Integrates deep inspection, intrusion prevention, app awareness.

Pros: Advanced threat protection

Cons: Expensive, can impact performance

III. Firewall Limitations

- Can't stop:
 - **Insider threats**
 - **Social engineering**
 - **Zero-day attacks**
- Can't inspect **encrypted traffic**
- Not a substitute for antivirus or endpoint protection

2. IDS, IPS, and Firewall Solutions

- **YARA Rules (Intrusion Detection based on YARA):**
 - A malware research tool that allows security analysts to detect and classify malware based on patterns (rules).
 - Rules define conditions (Boolean expressions) and strings (text, hex, regex) that indicate malware.
 - Tools: yarGen (generates YARA rules).
- **Intrusion Detection Tools:**
 - **Snort:** Open-source network intrusion detection system. Performs real-time traffic analysis, packet logging, and intrusion prevention. Uses rules to identify patterns.

- Uses of Snort Straight packet sniffer like tcpdump
 - Packet logger (useful for network traffic debugging, etc.)
 - Network intrusion prevention system
- **Suricata:** Open-source network IDS/IPS/NSM engine. Offers high performance, multi-threading.
- Other NIDS/HIDS: Juniper Network's IDP system, Samhain HIDS, OSSEC, Zeek, Cisco Secure IPS.
- **Intrusion Prevention Tools:**
 - **Trellix Intrusion Prevention System:** Detects and blocks advanced threats, worms, and reconnaissance attacks. It aggregates flow data from switches and routers and performs network-level behavior analysis
 - Other IPS: Check Point Quantum IPS, Atomic OSSEC, McAfee Host IPS, Secure IPS (NGIPS), Palo Alto Advanced Threat Prevention.
- **Firewall Solutions: Firewalls Based on Working Mechanism**
 - IPFire (<https://www.ipfire.org>)
 - pfSense (<https://www.pfsense.org>)
 - SonicWall TZ Series (<https://www.sonicwall.com>)
 - BIG-IP Advanced Firewall Manager (<https://www.f5.com>)

IDS/Firewall Evasion Techniques

- **Port Scanning:** Using specific ports (e.g., HTTP, HTTPS, DNS, FTP) that are allowed by firewalls to scan for vulnerabilities.
- **Firewalking:** Using TTL values to determine gateway ACL filters and it maps networks by analyzing the IP packet responses. irewalk and the Nmap firewalk script provide critical insights into network configurations and firewall rules, which can be leveraged to bypass security measures, including IDS
- **Banner Grabbing:** Obtaining service banners (vendor, version info) to identify devices and their configurations, which may reveal vulnerabilities. This knowledge allows attackers to understand how the IDS is set up and tailor their actions to avoid detection
- **IP Address Spoofing:** Masquerading as a trusted IP source to bypass IP-based filters. Tools such as Hping allow attackers to create custom packets with forged IP addresses
- **Source Routing:** Specifies the path of packets to bypass monitoring or less-monitored segments.
- **Tiny Fragments:** Sending tiny IP fragments that IDS struggles to reassemble, causing them to miss malicious patterns. This attack is used to avoid user defined filtering rules and works when the firewall checks only for the TCP header information
- **Using an IP Address in Place of a URL:** This method involves typing the IP address directly into the browser's address bar in place of typing the blocked website's domain name Use services such as Host2ip to find the IP address of the blocked website This method fails if the blocking software tracks the IP address sent to the web server

- **Using Anonymous Website Surfing Sites:** Using proxy servers or VPNs to hide real IP addresses.
- **Using a Proxy Server:** Directing traffic through a proxy to bypass local network filters.
- **ICMP Tunneling:** Encapsulating malicious commands/payloads within ICMP Echo packets to create a covert channel. The attackers use ICMPTX tool
- **ACK Tunneling:** Using ACK packets to carry data or commands through firewalls, as ACK packets are often less scrutinized. Using tools like Hping or Nping, the attacker crafts ACK packets containing malicious payload
- **HTTP Tunneling:** Encapsulating non-HTTP traffic (e.g., SSH, FTP) within HTTP requests (port 80/443) to bypass firewalls.
 - **HTTPPort and HTTPHost:** Tools to perform HTTP tunneling.
- **SSH Tunneling:** Creating an encrypted SSH tunnel to bypass firewall rules and access remote services (e.g., via OpenSSH or Bitvise)

Type	Description	Purpose / Use Case
Local Port Forwarding (LPF)	Forwards a local port to a remote server	Access internal services from outside
Remote Port Forwarding (RPF)	Opens a port on the remote server to forward to local machine	Allow access to local apps from the remote server
Dynamic Port Forwarding (DPF)	Creates a SOCKS proxy to route any traffic	Full proxy access, ideal for browsing or app tunneling

- **DNS Tunneling:** Encapsulating traffic within DNS queries and responses to create a covert channel.
 - **Iodine/dns2tcp:** Tools for DNS tunneling.
- **Bypassing through External Systems:** Compromising an external system (e.g., employee's home machine, third-party vendor) to launch attacks from a trusted source. Attackers bypass firewalls/IDS by hijacking session data from a legitimate user accessing the corporate network via an external system. They exploit the user's browser process to redirect it to a malicious server, where malware is downloaded and executed.
- **Bypassing through MITM Attacks:** Performing DNS server poisoning or routing manipulation to redirect traffic through the attacker's system.
- **Bypassing through Content:** Embedding malicious code within legitimate files (images, documents, audio/video) or disguising it using steganography or obfuscation.
- **Bypassing WAF (Web Application Firewall) using XSS Attack:** Exploiting XSS vulnerabilities to bypass WAFs by using ASCII values, Hex encoding, or obfuscation to hide malicious payloads.
 - **HTTP Header Spoofing:** Modifying HTTP headers to trick WAFs or bypass rules.

- **Blacklist Detection:** Identifying and bypassing WAFs that use blacklists by finding unlisted keywords.
- **Fuzzing/Brute-forcing:** Sending multiple payloads to find a bypass.
- **Abusing SSL/TLS ciphers:** Exploiting weak SSL/TLS cipher suites to bypass WAFs.
- **Bypassing IDS/Firewall through HTML Smuggling:** Embedding malicious code as a JavaScript Blob within an HTML attachment or web page. The client's browser downloads and executes the malicious payload.
- **Evading IDS/Firewall through Windows BITS (Background Intelligent Transfer Service):** Using BITS, a legitimate Windows service for file transfers, to download and execute malicious payloads, bypassing security solutions.

Other Techniques for IDS Evasion

- **Insertion Attack:** Attacker sends invalid packets that the IDS accepts but the destination system rejects, leading to desynchronization.
- **Evasion Attack:** Attacker sends packets that IDS discards but the host accepts, leading to desynchronization.
- **Denial-of-Service Attack (DoS):** Overwhelming IDS resources (memory, CPU) with noise traffic to prevent it from detecting true attacks.
- **Obfuscation:** Making code difficult to understand or detect (e.g., Unicode, ASCII, polymorphism, metamorphism).
- **False Positive Generation:** Constructing malicious packets to trigger numerous false alarms, distracting IDS administrators.
- **Session Splicing:** Exploiting IDS's inability to reconstruct sessions from fragmented packets.
- **Unicode Evasion:** Using Unicode characters to bypass signature-based detection.
- **Fragmentation Attack:** Breaking packets into small fragments that IDS fails to reassemble or incorrectly analyzes.
- **Time-to-Live (TTL) Attacks:** Manipulating TTL values to bypass IDS by making packets expire before reaching the IDS or router.
- **Urgency Flag:** Using the TCP urgency pointer to hide malicious data within normal data streams.
- **Invalid RST Packets:** Sending invalid RST packets to desynchronize communication.
- **Polymorphic Shellcode:** Mutating shellcode to evade signature detection.
- **ASCII Shellcode:** Using only ASCII characters for shellcode to bypass IDS filters.
- **Application-Layer Attacks:** Exploiting vulnerabilities in application layer protocols.
- **Desynchronization:** Breaking the synchronization between IDS and the target host (Pre-Connection SYN, Post-Connection SYN).

- **Domain Generation Algorithms (DGA):** Malicious software generates new domain names for C&C communication, evading static blacklists.
- **Encryption:** Using encrypted communication (SSH, SSL/TLS, VPN) to hide malicious traffic from IDS/firewalls.
- **Flooding:** Overloading IDS resources with traffic to bypass security.

4. Evading NAC and Endpoint Security

Network Access Control (NAC): Security control to block unauthorized users/hosts from internal services.

Endpoint Security: Protects end-user devices (desktops, laptops, mobiles, servers) from malware and other cyber threats.

NAC and Endpoint Security Evasion Techniques

- **VLAN Hopping:** Bypassing VLAN segmentation to access restricted networks.
- **Using a Pre-authenticated Device:** Using a device that is already authenticated to the network.
- **Ghostwriting:** Using fake credentials or bypassing authentication mechanisms.
- **CPL (Control Panel) Side-loading:** Exploiting legitimate Windows features.
- **Using Application Whitelisting:** Bypassing application whitelisting.
- **Dechaining Macros:** Obfuscating malicious macros.
- **Clearing Memory Hooks:** Removing traces of malware from memory.
- **Process Injection:** Injecting malicious code into legitimate processes.
- **Using LOLBins (Living Off the Land Binaries):** Using legitimate system binaries for malicious purposes.
- **Using ChatGPT:** For generating attack payloads or social engineering content.
- **Using Metasploit Templates:** Exploiting template vulnerabilities.
- **Windows Antimalware Scan Interface (AMSI) Bypass:** Techniques to evade AMSI detection.
- **Hosting Phishing Sites:** Hosting malicious sites for credential harvesting.
- **Reducing Entropy:** Reducing randomness in payloads to bypass detection.
- **Escaping the Sandbox:** Techniques to detect and evade sandbox environments.
- **Disabling Event Tracing for Windows:** Preventing security logging.
- **Spoofing "Admins at the Syn/Ack":** Mimicking legitimate network traffic.
- **Timing-Based Evasion:** Delaying execution to bypass time-based detection.
- **Signed Binary Proxy Execution:** Using legitimate signed binaries to proxy malicious code.
- **Bypassing the Thread Call Stack:** Modifying thread stack to evade detection.

- **Shared Memory Encryption:** Encrypting data in shared memory.
- **Summary/Encryption of Beacon:** Obfuscating beacon communication.

Traffic IQ Professional is a security auditing tool used to simulate application or attack traffic for testing firewalls, IDS/IPS, and other network devices. While designed for defensive assessments, attackers exploit it to craft custom traffic and bypass perimeter defenses.

5. Honeypot Concepts and Detection

What is a Honeypot?

A security resource designed to be attacked and exploited, serving as a trap for cyber attackers. It's a decoy system used to lure attackers, gather intelligence on their methods, and distract them from real production systems.

- **Purpose:**
 - Gain information about attacker methods/tools/motivations.
 - Divert attackers from critical systems.
 - Record detailed attack activities.
 - Identify new threats and vulnerabilities (zero-days).
 - Learn about attacker behavior in real-time.

Types of Honeypots

Classification of honeypots based on their design criteria:

Low-interaction Honeypots	These honeypots simulate only a limited number of services and applications of a target system or network
Medium-interaction Honeypots	These honeypots simulate a real operating system , applications, and services of a target network
High-interaction Honeypots	These honeypots simulate all services and applications of a target network
Pure Honeypots	These honeypots emulate the real production network of a target organization

Classification of honeypots based on their deployment strategy:

- Production Honeypots
- Research Honeypots

Classification of honeypots based on their deception technology:

- Malware Honeypots
- Spam Honeypots
- Spider Honeypots
- Database Honeypots
- Email Honeypots
- Honeynets

Honeypot Tools

Honeybots

HoneyBOT is an easy-to-use solution that is ideal for network security research or as part of an early-warning IDS

Malware honeypots: Malware honeypots are used to trap malware campaigns or malware attempts over the network infrastructure. These honeypots are simulated with known vulnerabilities such as outdated APIs, vulnerable SMBv1 protocols, etc., and they also emulate different Trojans, viruses, and backdoors that encourage adversaries to perform exploitation activities

Spam honeypots: Spam honeypots specifically target spammers who abuse vulnerable resources such as open mail relays and open proxies. Basically, spam honeypots consist of mail servers that deliberately accept emails from any random source from the Internet

Spider honeypots: Spider honeypots are also called spider traps. These honeypots are specifically designed to trap web crawlers and spiders. Many threat actors perform web crawling and spidering to extract important information from web applications. Such crucial information includes URLs, contact details, directory details, etc

Honeynets: Honeynets are networks of honeypots. They are very effective in determining the entire capabilities of the adversaries. Honeynets are mostly deployed in an isolated virtual environment along with a combination of vulnerable servers. The various TTPs employed by different attackers to enumerate and exploit networks will be recorded, and this information can be very effective in determining the complete capabilities of the adversary

Detecting Honeypots

Fingerprinting the Running Service	• Method: Identify specific services on network using nmap tool • Example Command: <code>nmap -sV -p 80 <target_ip></code> • Indicators: Discrepancies between claimed and actual service behaviors
Analyzing Response Time	• Method: Measure latency of responses using tools such as Ping, Traceroute or nmap • Example Command: <code>nmap -p 80 --scan-delay 1s --max-retries 5 <target_ip></code> • Indicators: Consistently higher response times and variability
Analyzing MAC Address	• Method: Examine MAC addresses to identify irregularities using tools such as Nmap or arp-scan • Example Command: <code>arp-scan --interface=eth0 --localnet</code> • Indicators: MAC addresses with unusual Organizationally Unique Identifiers (OUI)
Enumerating Unexpected Open Ports	• Method: Identify unusual or unexpected open ports using tools such as Nmap • Example Command: <code>nmap -p- <target_ip></code> • Indicators: Open ports that do not align with expected services
Analyzing System Configuration and Metadata	• Method: Scrutinize configuration settings, system banners, and metadata for inconsistencies • Indicators: Default configurations, outdated banners, and discrepancies in system information

Honeypot Detection Tools

Attackers use honeypot detection tools such as Send-Safe Honeypot and SniffingBear to detect honeypots in the target organizational networks. ▪ Send-Safe Honeypot Hunter Send-Safe Honeypot Hunter is a tool designed for checking lists of HTTPS and SOCKS proxies for "honey pot"

6. IDS/Firewall Evasion Countermeasures

- **Regular Updates:** Keep IDS/IPS/Firewall software and signatures up-to-date.
- **Strong Policies:** Implement strict security policies and access controls.
- **Layered Defense:** Deploy multiple security layers (firewall, IDS, IPS, EDR, NAC).
- **Encryption:** Use strong encryption for all sensitive communication (SSL/TLS, IPsec, VPN).
- **Traffic Analysis & Baselines:** Continuously monitor network traffic for anomalies and deviations from normal baselines.
- **Network Segmentation:** Divide the network into smaller segments to limit the scope of an attack.
- **Egress Filtering:** Filter outbound traffic to prevent IP spoofing and unauthorized connections.
- **Ingress Filtering:** Filter inbound traffic to ensure legitimate source IPs.
- **TCP Intercept/SYN Cookies:** Protect against SYN flood attacks.
- **Rate Limiting:** Limit the number of connections or requests from a single source.
- **Unified Threat Management (UTM) / Next-Gen Firewalls (NGFW):** Integrated security solutions with advanced features.
- **Anti-Phishing & Anti-Malware:** Prevent initial infection vectors.
- **Authentication & Authorization:** Strong authentication mechanisms (MFA) and proper access privileges.

- **Patch Management:** Regularly patch all systems and applications.
- **User Awareness Training:** Educate users about evasion techniques and social engineering.
- **Web Development Guidelines:** Secure session ID generation, HSTS, secure cookies.
- **DNSSEC:** Secure DNS infrastructure.
- **Traffic Normalization:** Normalize traffic to remove ambiguities that attackers exploit.
- **Behavioral Biometrics:** Use typing rhythm, mouse movements for continuous authentication.
- **Challenge-Response Mechanisms:** CAPTCHA to detect suspicious activity.
- **Session Timeout:** Automatically terminate inactive sessions.
- **Centralized Logging & SIEM:** Aggregate logs for correlation and anomaly detection.
- **Threat Intelligence Feeds:** Integrate threat intelligence to identify new evasion techniques.
- **DDoS Protection Services/Appliances:** To mitigate volumetric attacks.

Key Terms for Last-Minute Revision

- **IDS (Intrusion Detection System):** Detects suspicious network/system activity.
- **IPS (Intrusion Prevention System):** Detects and *prevents* suspicious activity.
- **Firewall:** Filters network traffic based on rules.
- **Signature Recognition (IDS):** Detects known attack patterns.
- **Anomaly Detection (IDS):** Detects deviations from normal behavior.
- **Protocol Anomaly Detection (IDS):** Detects deviations from protocol standards.
- **NIDS (Network-Based IDS):** Monitors network traffic.
- **HIDS (Host-Based IDS):** Monitors individual hosts.
- **True Positive:** Correctly identified attack.
- **False Positive:** Incorrectly identified attack.
- **False Negative:** Missed attack (most critical).
- **Bastion Host:** Highly secured gateway server.
- **DMZ (Demilitarized Zone):** Perimeter network segment.
- **Packet Filtering:** Basic firewall filtering based on headers.
- **Circuit-Level Gateway:** Firewall operating at Session layer.
- **Application-Level Firewall (Proxy Firewall):** Filters at Application layer.
- **Stateful Multilayer Inspection:** Firewall that tracks connection state across layers.

- **NAT (Network Address Translation):** Hides internal IPs.
- **VPN (Virtual Private Network):** Creates secure, encrypted tunnel.
- **NGFW (Next-Generation Firewall):** Advanced, integrated firewall.
- **YARA Rules:** Pattern-matching for malware detection.
- **Snort:** Popular open-source NIDS/NIPS.
- **Suricata:** High-performance NIDS/NIPS.
- **Port Scanning:** Identifying open ports on a target.
- **Firewalking:** Mapping firewall rulesets.
- **Banner Grabbing:** Obtaining service version info.
- **IP Spoofing:** Forging source IP addresses.
- **Source Routing:** Specifying packet path.
- **Tiny Fragments:** Breaking packets to evade detection.
- **ICMP Tunneling:** Encapsulating data in ICMP packets.
- **ACK Tunneling:** Using ACK packets to carry data.
- **HTTP Tunneling:** Encapsulating traffic within HTTP requests.
- **SSH Tunneling:** Creating an encrypted tunnel over SSH.
- **DNS Tunneling:** Encapsulating data within DNS queries.
- **HTML Smuggling:** Embedding malicious code as a JavaScript Blob in HTML.
- **Windows BITS:** Using Background Intelligent Transfer Service for malware transfer.
- **Obfuscation:** Making code difficult to understand or detect.
- **Polymorphic Shellcode:** Mutating shellcode.
- **ASCII Shellcode:** Shellcode using only ASCII characters.
- **Fragmentation Attack (IDS):** Overloading IDS with fragmented packets.
- **TTL (Time-to-Live) Attacks:** Manipulating TTL to bypass IDS/routers.
- **Urgency Flag:** Using TCP urgency pointer to hide data.
- **Desynchronization (IDS):** Breaking synchronization between IDS and target.
- **DGA (Domain Generation Algorithm):** Generates random domain names for C&C.
- **NAC (Network Access Control):** Controls access to network based on policy.
- **Endpoint Security:** Protects individual devices.
- **LOLBins (Living Off the Land Binaries):** Using legitimate system tools for malicious purposes.
- **Honeypot:** Decoy system to lure and study attackers.

- **Low-interaction Honeypot:** Simulates limited services.
- **High-interaction Honeypot:** Full system emulation.
- **HSTS (HTTP Strict Transport Security):** Forces HTTPS.
- **Token Binding:** Binds tokens to TLS session.
- **DNS over HTTPS (DoH):** Encrypts DNS queries.
- **IPsec:** Suite of protocols for secure IP communication.
- **SIEM (Security Information and Event Management):** Centralized logging and analysis.

Layer 4 Tar Pits: Layer 4 tar pits manipulate the TCP/IP stack and are effectively employed to slow down the spreading of worms, backdoors, etc. In these tar pits, the iptables accept the incoming TCP/IP connection and spontaneously switch to a zero-window size, blocking the attacker from sending further data

Layer 2 Tar Pits: Layer 2 tar pits are used to block the network penetration of the attacker who gains access to the network as well as to prevent internal threats. The attacker can detect the presence of this daemon by looking at the responses with the unique MAC address 0:0:f:ff:ff:ff, which acts as a kind of black hole

Layer 7 Tar Pits: Layer 7 tar pits react slowly to incoming SMTP commands by attackers/ spammers. Attackers can identify the presence of Layer 7 tar pits by looking at the latency of the response from the service

Honeyd: Honeyd is a widely used honeypot daemon. It is used to create thousands of honeypots easily. It is a network-simulated and service-simulated honeypot deployment engine. This honeyd honeypot can respond to a remote attacker who tries to contact the SMTP service with fake responses

One of the following techniques redirects all malicious network traffic to a honeypot after any intrusion attempt is detected. Attackers can identify such honeypots by examining specific TCP/IP parameters such as the round-trip time (RTT), time to live (TTL), and TCP timestamp. Which is this technique?

Bait and switch